Non-local GNNs for Jet Classification

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About Me

I'm an undergraduate Electrical Engineering Student at Delhi Technological University, India. I'm interested in machine learning and artificial intelligence and have worked on projects like an AI-Powered Article Summarization Platform that utilizes hybrid summarization pipelines combining both extractive and abstractive models and Space debris detection and optimization using YOLOv8 architecture and implementing fine-tuning techniques to enhance model accuracies.

I was part of Samsung Innovation Lab at DTU where I worked on a research project regarding dementia classification and presented the work at a AAIC Neuroscience Next India conference. I have completed all three GENIE evaluation tasks which are available on my GitHub repository. I'm proficient in Python, TensorFlow and PyTorch which makes me well suited for this organisation.

Mentors:

- Sergei Gleyzer (University of Alabama)
- Ali Hariri (EPFL)
- Tom Magorsch (TUM)

Synopsis

This project builds on previous projects by leveraging non-local graph neural networks (GNNs) for the classification of quarks and gluons in jets at the LHC. This approach uses the inherent structure among particles within a jet by representing them as a graph instead of treating them like independent entities like in the case of convolutional autoencoders. Multiple different architectures such as EdgeConv + self-attention and graph transformer will be implemented and compared to a local EdgeConv baseline. Evaluation will be done using ROC-AUC on the quark/gluon jet dataset.

Why ML4SCI

ML4SCI stands out to me among most GSoC organisations as it primarily focuses on research instead of commercial applications. I am also interested in applying novel ML techniques to improve the understanding of science especially in the field of particle physics.

The opportunity to work under these mentors and researchers on real world problems is also really fascinating.

Project Description

Jet Classification at LHC is essential as it allows to study them for physics and bad jet classification can lead to biased physics results. Jets are naturally sparse as most pixels are zeros, so representing them as graphs is more natural. Current GNN approaches use local neural networks which leads them to miss the long range dependencies inherent to jets.

Non-local GNNs fixes this by adding self-attention layers which allows every particle to directly attend to other particles in the jet. We compared the local variant baseline to non-local variants on 20k events. The evaluation tasks shows that all models achieve 0.78-0.80 AUC on these events. EdgeConv achieved 0.79 AUC while Graph Transformer achieved 0.794 AUC

During GSoC, these models will be scaled to the full 139k dataset and we will add edge features, positional encodings and dynamic graph recomputation to better improve the results.

Technical Approach

Data Representation

- Images (125x125x3) were converted to point clouds after the non-zero pixels were extracted
- Each node was given Log-transformed ECAL, HCAL, Track energies, Total energy, η and φ coordinates as features
- The graphs were constructed using k=7 nearest neighbours
- Features were standardized and normalized

Architecture

1. Baseline (Local)

- 3 layer EdgeConv was selected as the baseline model
- Each layer was given a shared MLP to pairs of neighbouring nodes
- Global mean and max pooling were used for graph representation

2. EdgeConv + Self-Attention

- The previous EdgeConv model was used as baseline
- After each EdgeConv layer, a self-attention multi-headed block was added which let every node attend to other nodes in the jet
- This led us to isolate the effect of non-locality by keeping everything else same

3. Graph Transformer

- TransformerConv layers were used for message passing

- Attention weights were learned which allowed the model to decide which neighbours mattered the most

Training Strategy

- AdamW optimizer was used along with gradient clipping and label smoothing
- Cross-validation was used for robust results
- ROC-AUC was used as primary metric along with accuracy, F1 and confusion matrices

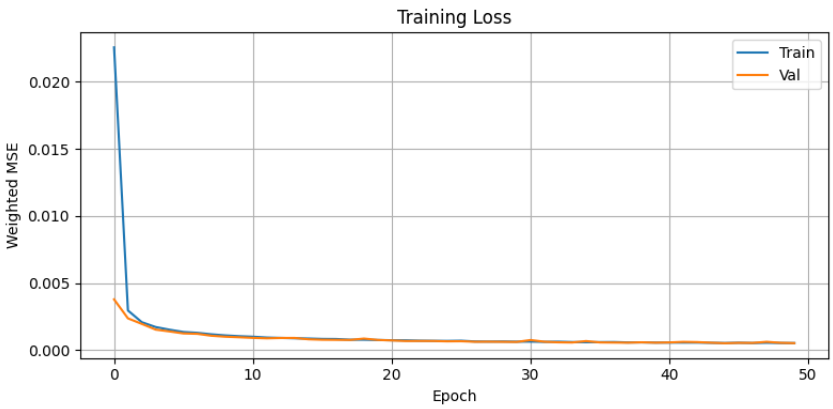
Evaluation tasks

Common task 1: Autoencoder for Quark/Gluon Jet Events

We trained a standard convolutional autoencoder on the quark/gluon jet dataset (139,306 events, 125x125x3 channels: ECAL, HCAL, Tracks). Only 10k events were utilized in this project due to RAM constraints on Google Colab. The key challenge in this was the extreme sparsity where only 0.5-2.7% of pixels were non-zero across channels.

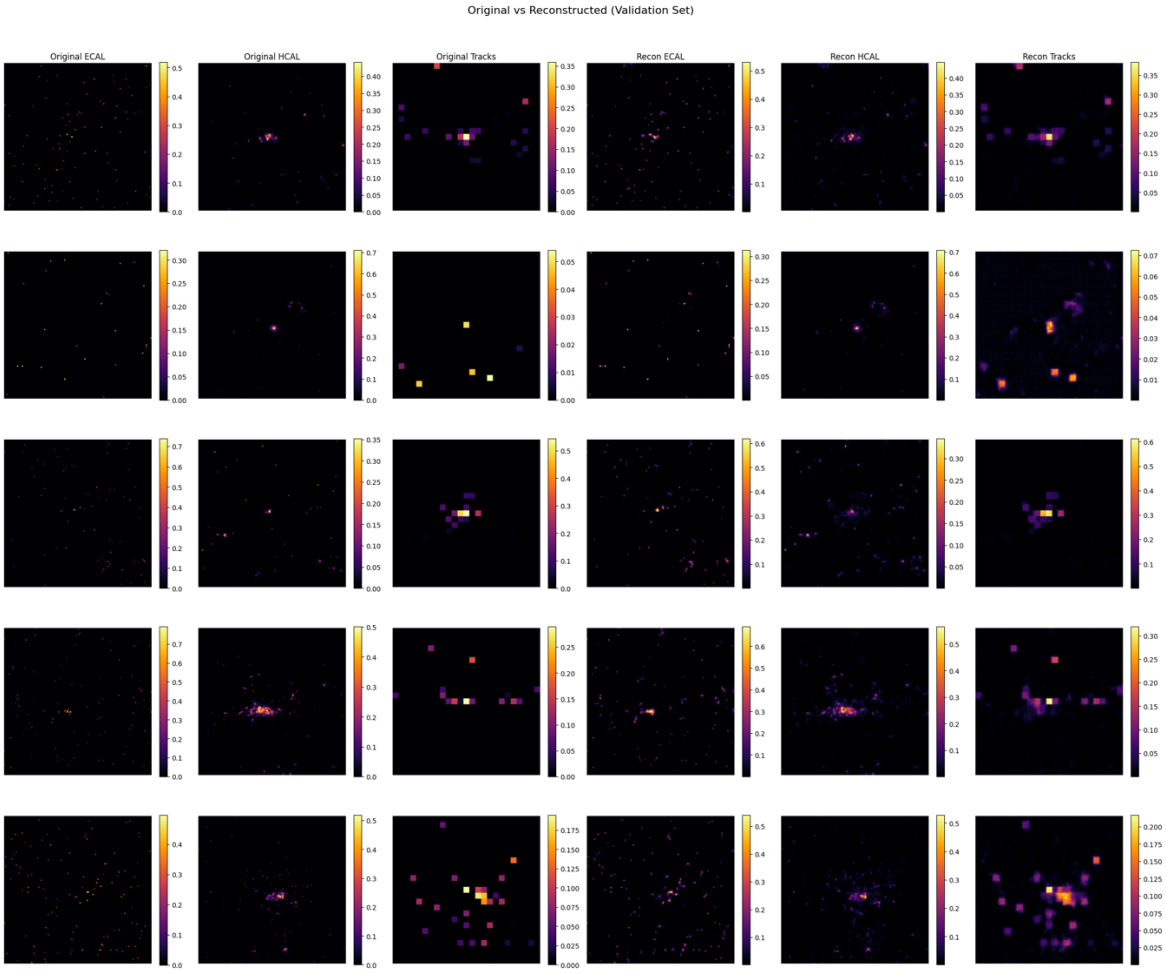
At first Standard MSE loss were used which failed as the model learned to predict zeros everywhere and produced a generic center blob as the reconstructed image. To fix this log transform (log1p(x × 1000) were used to compress the dynamic range and spread out the extremely small energy values and a weighted MSE loss was used that penalizes errors on nonzero pixels 50x more heavily. Per-channel normalization was also applied as each channel has different scales (ECAL max = 10, Tracks max = 0.16)

A fully convolutional autoencoder with no FC bottleneck was used which preserved the spatial information through latent space (16x16x32 feature map). The encoder had 3 conv layers with stride 2, followed by a 1×1 channel compression layer. The decoder had symmetric transposed convolutions with outputs cropped from 128 to 125. The total parameters in the model were 199,907.



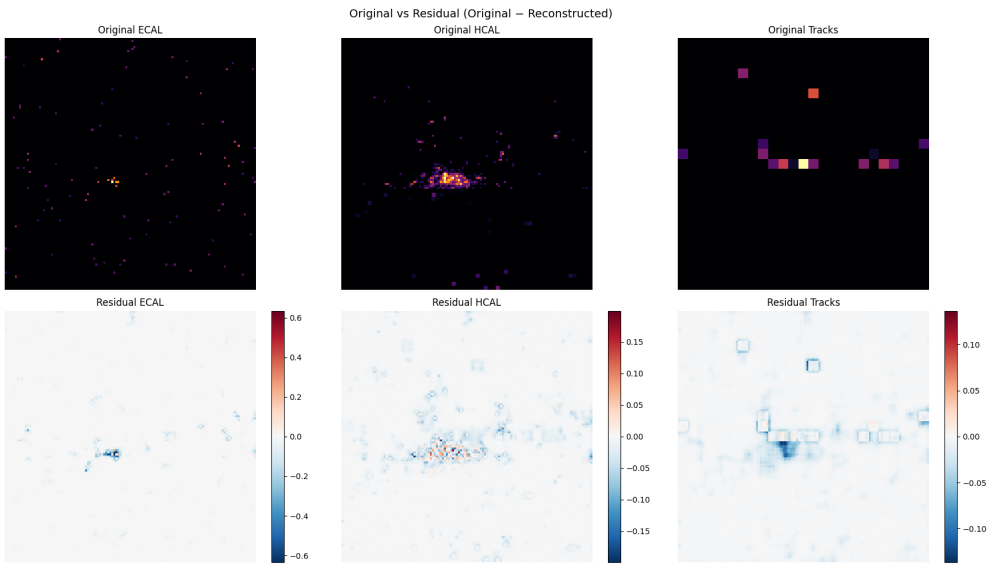
Training loss curve

The model was trained for 50 epochs with Adam optimizer and lr=1e-3 with ReduceLROnPlateau scheduler. The reconstruction captured the spatial structure of energy across all the channels.



Original vs Reconstructed images from the validation set

The residual plots showed that the errors were concentrated at energy deposit locations and not random noise. The latent space shows partial clustering by jet type without labels which shows that the model captured physically meaningful features of the jet substructure.



Residual plots of the Original vs Reconstructed images

Common Task 2: GNN for Jet Classification

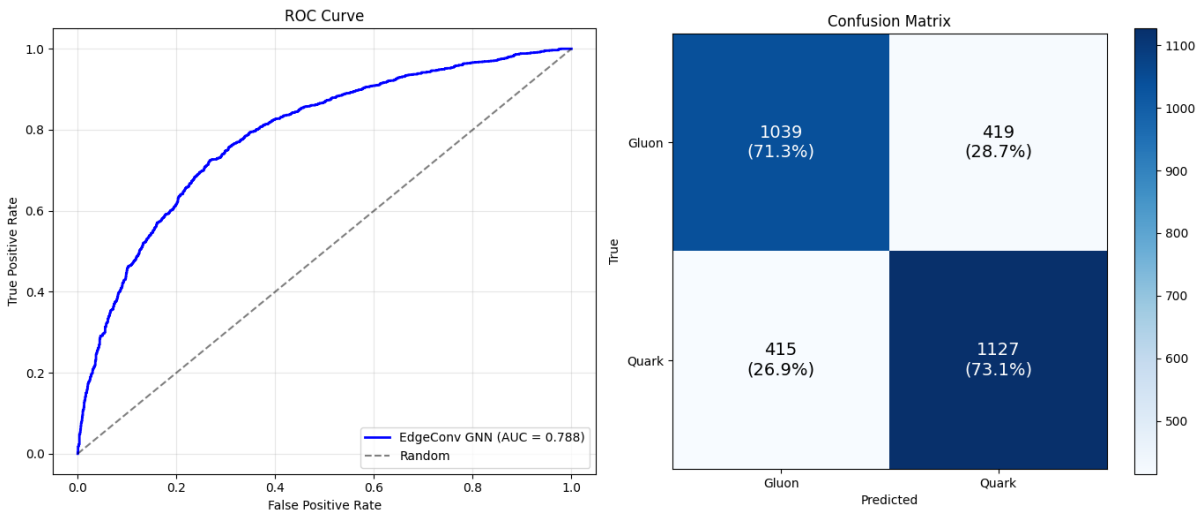
We built a 3-layer EdgeConv GNN with max aggregation for binary quark/gluon classification which was inspired by the ParticleNet architecture. Each jet image was converted into a point cloud by extracting all the non-zero pixels. Only 20k Events were used and the graphs were capped at 150 nodes in which the highest energy hits were retained. Each nonzero pixel became a node with 6 features η, φ coordinates and log-transformed ECAL, HCAL, Track, and total energies. Graphs were constructed using k=7 nearest neighbours in (η, φ) space.

Each EdgeConv block applies a shared MLP to pairs of neighbouring node features and aggregates via max pooling. Graph-level representation is obtained by concatenating global mean and max pooling over all nodes.

```
Test Accuracy: 0.7220
Test AUC:      0.7885

              precision    recall  f1-score   support

       Gluon       0.71      0.71      0.71      1458
       Quark       0.73      0.73      0.73      1542

    accuracy                           0.72      3000
   macro avg       0.72      0.72      0.72      3000
weighted avg       0.72      0.72      0.72      3000
```

Results table

The Model achieved a test AUC of 0.7885 and an accuracy of 72% which are consistent with the usual GNN baselines on this dataset.



AUC score and Confusion Matrix for the EdgeConv Model

Specific Task 4: Non-local GNNs

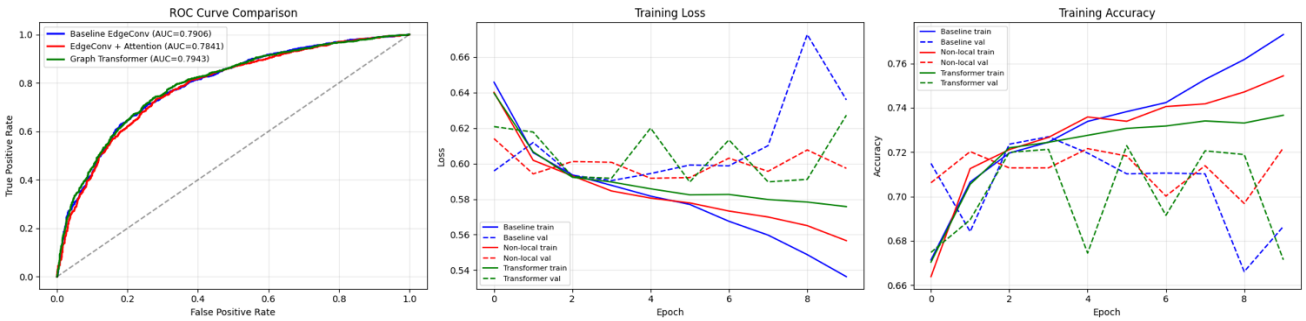
In this task, we compared the performances of local and non-local graph neural networks for the classification of quark/gluon jets. Three architectures were evaluated after training them on identical data.

1. Baseline EdgeConv – local message passing only
2. EdgeConv + Self-Attention – local message passing augmented with global self-attention after each layer
3. Graph Transformer – multi-head attention-based message passing

Each 125x125x3 jet image was converted into a graph as it was in the common tasks. The non-zero pixels (Nodes) were capped at 150 to retain the highest energy hits. Each node had 6 features: η, φ coordinates + log-transformed ECAL, HCAL, Track, and total energy and k=7 nearest neighbours.

For the EdgeConv + Self-Attention model, the same EdgeConv backbone was used but after each messaging passing layer, a self-attention block was added that introduced non-local information flow while also preserving the local bias from EdgeConv. For the Graph Transformer, the EdgeConv was entirely replaced with TransformerConv which uses multi-head attention for message passing along edges.

All three models were trained with identical settings: AdamW optimizer (lr=5e-4, weight decay=1e-2), cross-entropy loss with label smoothing (0.1), gradient clipping at 0.5, for 10 epochs.



Model	AUC	Acc	Params
Baseline EdgeConv	0.7906	72.2%	319,938
EdgeConv + Attention	0.7841	72.4%	666,690
Graph Transformer	0.7943	72.3%	381,442

Summary of Results of the 3 models

All three models had comparable performances (AUC of 0.7841-0.7943) with the Graph Transformer achieving the highest AUC. The Graph Transformer which uses non-local message passing slightly outperforms the baseline local EdgeConv by AUC. However, the EdgeConv + Self-Attention hybrid did not consistently outperform the baseline which could be because the added global attention introduces noise that dilutes the attention signal.

Deliverables

- Full data pipeline for the complete 139K event dataset that loads and preprocesses data efficiently
- Baseline EdgeConv trained and benchmarked on the full dataset
- EdgeConv + Self-Attention non local model trained and benchmarked
- Graph Transformer model trained and benchmarked
- GPSConv implementation (local MPNN + global attention combined)
- Report comparing all architectures with ROC-AUC, accuracy, F1 scores, confusion matrices.
- Clean and documented open-source codebase on GitHub
- Blog post summarizing all findings

Timeline

Week 1: Literature review on non-local GNNS and their application in particle physics. Study ParticleNet, Graph Transformers, GPSConv and recent papers to understand the current landscape.

Week 2: Build the full data pipeline for the entire dataset and implement efficient batched loading, image to point conversion, feature normalization, k-nn graph construction and ensuring that the pipeline can handle memory constraints properly.

Week 3: Train the baseline EdgeConv model and establish a benchmark with the AUC numbers with proper train/val/test splits. Document baseline results.

Week 4: Implement and train the EdgeConv + Self-Attention hybrid on the full dataset. Compare against baseline using ROC-AUC and other metrics. Analyse the effects of adding non-local attention to the baseline model.

Week 5: Implement and train the Graph Transformer on the full dataset. Compare all three architectures and prepare a mid-project report summarizing all progress and findings for mentor feedback.

Week 6: Implement GPSConv which is the current state-of-the-art graph transformer architecture and evaluate against previous models.

Week 7: Add edge features and positional encodings to the best performing models and Evaluate the impact of adding these features on classification performance.

Week 8: Explore DynamicEdgeConv for graph Recomputation in learned feature space.

Week 9: Hyperparameter tuning for all models and running cross-validation to ensure correct and robust results.

Week 10: Final and Comprehensive benchmarking. Run each model with multiple seeds for statistical significance. Analyse where non-local models impact most.

Week 11: Complete documentation for the full codebase and Prepare the blog post with the benchmark reports.

Week 12: Clean up the repository and ensure that the code is reproducible. Prepare for final evaluation and Discuss potential post-GSoC contributions with mentors.

Related Work

The baseline architecture is inspired by ParticleNet which first applied the EdgeConv to jet tagging and showed that treating jets as points clouds worked better than image-based approaches. GPSConv showed that combining both local message passing with global self-attention in each layer gives the best of both worlds and is considered sota model for graph-level tasks. On the physics side, LorentzNet introduced Lorentz-equivariant message passing which is a possible future extension of this work.

We directly compare local vs non-local GNNs on the same jet dataset in this project and plan to extend towards architectures like GPSConv during GSoC. The Work also builds on the success of graph-based models used in previous GSoC projects under GENIE at ML4SCI.

Benefits to Community

This project will produce an open-source benchmark comparing local and non-local GNN architectures for jet classification that other researchers and future GSoC contributors can build on. The results will help guide architecture choices for graph-based tasks in particle physics and the methods generalize to other point cloud classification problems in science.

References

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